Ali Golestaneh

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EDUCATION

Worcester Polytechnic Institute (WPI) — *Aug 2024 – Present*
Doctorate of Philosophy, Robotics Engineering — *Worcester, MA*
Total GPA via first 36 credits: **4/4**

Iran University of Science and Technology (IUST) — *Sep 2018 – Sep 2023*
Bachelor of Science, Mechanical Engineering — *Tehran, Iran*
Total GPA: **16.12/20 (3.34/4)**

National Organization for Development of Exceptional Talents (NODET) — *Sep 2014 – Jul 2018*
High School Diploma in Mathematics and Physics — *Tehran, Iran*
Total GPA: **19.01/20 (4/4)**

HONORS AND AWARDS

- **Best Student Paper Award** at IEEE International Conference on Robotics and Automation (ICRA) 2026.
- **Best Student Paper Award** at Hellenic Robotics Forum (HRF) 2026.
- **Glenn Yee 2026 Travel Award** of Robotics Engineering Graduate Student.
- **Top 10%** within Mechanical Engineering students class of 2022.
- **Top 0.75%** among 144437 students in the National University Entrance Exam 2018.
- **Ranked 4th** at Robocup 2014 Iran Open International Competitions (Soccer 2D Simulation).

PUBLICATIONS

Golestaneh, S., Zhong, Z., Lee, D., Chamzas, C., **AURA: Asymptotically Optimal Uncertainty-Robust Replanning Algorithm for Kinodynamic Systems** [IEEE Robotics and Automation Letters] **Link**

Zhong, Z., Golestaneh, S., Chamzas, C., **ActivePusher: Active Learning and Planning with Residual Physics for Nonprehensile Manipulation**, *ICRA 2026* [Best Student Paper at ICRA 2026 and HRF 2026] **Link**

Yazdipaz, K., Kohli, N., Golestaneh, SA., Shahbazi, M., **Robust and Efficient Phase Estimation in Legged Robots via Signal Imaging and Deep Neural Networks**, *IEEE Access* **Link**

Zhong, Z., Golestaneh, S., Chamzas, C., **Terminal Matters: Kinodynamic Planning with a Terminal Cost and Learned Uncertainty in Belief State-Cost Space** [Under Review at IEEE Transactions on Robotics] **Link**

Golestaneh, S.*, Lee, D.*, Singh, J., Zhong, Z., Chamzas, C., **MetaPusher: Meta Learning and Planning for Nonprehensile Manipulation of Unseen Objects with Rapid Online Adaptation** [Under Review at IEEE International Conference on Robotics and Automation] **Link**

SKILLS

Programming Languages: C++, Python, Matlab, EES

Engineering Software: MuJoCo, OMPL, Isaac Gym, PyTorch, UR RTDE, SolidWorks, CATIA, ADAMS, Arduino

Operating Systems: Linux, ROS, Windows

Publishing Tools: LaTeX, Microsoft Office Package, Vim

Languages: Persian (Native), English (Fluent)

- **IELTS: 7 → L: 8.5 R: 6.5 W: 6 S: 6.5**
- **GRE: 312 → V:142 Q:170**

PROJECTS

Ph.D. Directed Research *Aug 2024 – Present*
Efficient Motion Planning for Kinodynamic Systems and Nonprehensile Manipulation *Worcester, MA*

- Extensive review of geometric and kinodynamic motion planners.
- Enhancing the efficiency of motion planning algorithms to maximize the accuracy.
- Implementing UR10/UR16e robot trajectory control methods in different simulation environments.
- Training Physics Residual Neural Networks to boost dynamics model efficiency with limited data.

Humanoid Robotics Development Project *Jan 2026*
Unitree G1 Development and Control *Worcester, MA*

- Developing and testing locomotion gait modes for the Unitree G1 humanoid robot.
- Implementing whole body motion control for stability.
- Integrating robot control commands within the Unitree G1 software framework and physical robot.

Robotics Control Project *Nov 2024*
Control and Plan for OpenManipulatorX *Worcester, MA*

- Implementing ROS-based forward and inverse kinematics of the robot arm.
- Applying kinematic velocity control through Jacobian inversion.
- Performing optimization of PD and torque controllers through gain calibration.

Bachelor's Dissertation *Oct 2021 – Sep 2023*
Analysis and Implementation of a Quadruped Robot Simulator *Tehran, Iran*

- Comprehensive study of legged robot gait algorithms and robotic arm dynamics.
- Building and testing MIT Mini Cheetah simulation and MiLAB robot simulation using Qt interface and ROS.
- Developing IUST quadruped robot simulator based on similar simulation environments.
- Adding a new joint controller for debugging the initial quadruped robot tests.

Mechanical Engineering Internship *Jul 2020 – Sep 2020*
Rahe Andisheh Company *Tehran, Iran*

- Developing a shrimp pond oxygen machine using reverse engineering.
- Redesigning the chassis design using SolidWorks to meet the company's new needs.
- Gearbox and actuator selection based on machine weight distribution.
- Checking structural features along with manufacturing process and testing.

WORK EXPERIENCE

Research Assistant

- *Learning Dynamics for Manipulation and Motion Planning* *Aug 2024 – Present*
  Supervisor: Dr. C. Chamzas, ELPIS Laboratory at WPI *Worcester, MA*
- *Simulation and Construction of Legged Robots* *Sep 2021 – Jul 2024*
  Supervisor: Dr. M. Shahbazi, Mechatronics Laboratory at IUST *Tehran, Iran*

Teaching Assistant

- *Motion Planning and Machine Learning for Robotics* *Aug 2025 – May 2026*
  Supervisors: Dr. C. Chamzas *Worcester, MA*
- *Mechanical Applications in Robotics and Sensing and Perception in Robotics* *Jan 2025 – May 2025*
  Supervisors: Dr. A. Rosendo and Dr. G. Lewin *Worcester, MA*
- *Fundamentals of Computer Programming* *Sep 2023 – Feb 2024*
  Supervisor: Dr. M. Shahbazi *Tehran, Iran*
- *Vectorial Dynamics and Dynamics of Machinery* *Sep 2022 – Feb 2023*
  Supervisors: Dr. S. Khanmirza and Dr. S. Asiaei *Tehran, Iran*

High School Teacher *Sep 2021 – May 2024*
Physics Teacher at National Organization for Development of Exceptional Talents High School *Tehran, Iran*